Separable spectral unmixing based on the learning of periodic absorbance changes: Application to functional brain mapping using RGB imaging

Charly Caredda[a], Jérémy E. Cohen[a], Laurent Mahieu-Williame[a], Raphaël Sablong[a], Michaël Sdika[a], Jacques Guyotat[c], and Bruno Montcel[a]

[a]Univ Lyon, INSA-Lyon, Université Claude Bernard Lyon 1, UJM-Saint Etienne, CNRS, Inserm, CREATIS UMR 5220, U1294, F69100, Lyon, France
[b]Service de Radiologie, Centre Hospitalier Universitaire de Saint Etienne, TAPE EA7423, Université de Lyon, UJM Saint Etienne, F42023 France
[c]Service de Neurochirurgie D, Hospices Civils de Lyon, Bron, France

ABSTRACT

Separable spectral unmixing designates techniques that allow to decompose spectra as a linear or non-linear combination of spectra of the targets (endmembers) collected. These techniques allow quantitative measurements but several drawbacks limit its use with standard optical devices like RGB cameras. We propose a new method for estimating endmembers and their proportion without calibration of the acquisition device with the analysis of periodic events in the signal. We evaluated the performances of the method for identifying functional brain areas during neurosurgery using RGB imaging. Results were consistent with clinical gold standards. This work can allow a widespread use of spectral imaging in the industrial or medical field.

Keywords: Spectral unmixing; auto-calibration; hemodynamic monitoring; functional brain mapping; RGB camera, optical imaging

1. INTRODUCTION

Separable spectral unmixing is a technique that was initially developed for remote sensing applications. Spectra recorded in satellite images are represented as a sum of all spectra collected by the sensor. This implies that the measured spectra are defined as a mixture of several constituent spectra (or endmembers), weighted by their proportion in the pixel (also called abundance).[1] Spectral unmixing is also used in biomedical applications, especially for *in vivo* optical imaging.[2] The objective is to observe non-invasively biological tissues, to help in understanding, detecting or tracking metabolic, functional or disease processes in the body. In a previous study, we implemented a method to quantify hemodynamics in patient brains using RGB imaging and the differential form of the modified Beer-Lambert law.[3] This method extracts the concentration changes of oxy- and deoxyhemoglobin by unmixing the measured absorbance changes in a linear combination of calibrated hemoblobin extinction spectra (endmembers).

Although spectral unmixing techniques allow quantitative measurements, several drawbacks limit its use with standard optical devices like RGB cameras. First, the devices must be calibrated, sometimes in complex environments such as operating rooms. Moreover, for some models, a *a priori* on the observed scene is also necessary. Without considering these two aspects, quantification of concentrations or abundances contains large estimation errors. In this paper, we introduce a new method called auto-calibration method for estimating endmembers and their proportion automatically and without calibration of the acquisition device. Endmembers are estimated using a separable non-negative matrix factorization (NMF).[4] Unlike traditional approaches, this method estimates the endmembers in spectral bands where absorbance changes periodically. The idea is to take advantage of the periodic temporal variations of the endmembers in order to estimate them automatically.

Further author information: (Send correspondence to C.C and B.M)
C.C.: E-mail: charly.caredda@creatis.insa-lyon.fr, B.M.: E-mail: bruno.montcel@creatis.insa-lyon.fr

We applied the auto-calibration method to a clinical application: the intraoperative functional brain mapping using an RGB camera. Non-invasive functional brain mapping is an imaging technique used to localize the functional areas of the patient brain. This technique is used during brain tumour resection surgery to indicate to the neurosurgeon the cortical tissues that should not be removed without cognitive impairment. Functional brain areas are then identified with statistical parametric mapping techniques.[5,6]

2. MATERIAL AND METHODS

2.1 Patient inclusion and experimental paradigm

Optical data were acquired during neurosurgery using RGB imaging with the wide field optical device described in[3] after the patient's craniotomy and before the brain tumor resection surgery. Optical data were acquired with successive periods of $20s$ of rest and $20s$ of motor stimulation that was repeated 3 times. Motor cortex stimulation was performed by the patient with repetitive, alternating hand opening and closing at ≈ 1 Hz. The neurosurgeon also performed electrical brain stimulation to identify functional brain areas.

2.2 Modified Beer Lambert law

Let us assume that the absorbance A can be expressed with modified Beer-Lambert law:

$$A(\lambda,t) = \log\left(\frac{I_0(\lambda)}{I(\lambda,t)}\right) = \mu_a(\lambda,t).L\left(\mu_a,\mu_s\right) + G(\lambda,t), \tag{1}$$

where I is the transmitted intensity, I_0, the input light intensity, μ_a and μ_s (in cm^{-1}), the absorption and scattering coefficients. G is an unknown geometry dependent factor whose values are not related to the variations of μ_a (for example, tissue dessication in biomedical applications). $L(\mu_a,\mu_s)$ (in cm) is the optical path length of the traveled photons dependent upon μ_a and μ_s.[7] Absorbance A can be seen to be a non-linear function of μ_a, the greatest deviation from linearity occurring at low μ_a and high μ_s values.[7] In a lot of applications, it is interesting to measure absorption changes. For this reasons the differential form of the modified Beer-Lambert law is generally computed:

$$\Delta A(\lambda,t) = \log\left(\frac{I_{ref}(\lambda)}{I(\lambda,t)}\right) = \Delta\mu_a(\lambda,t).L\left(\mu_a^{ref},\mu_s^{ref}\right) + \mu_a(\lambda,t).\Delta L(\lambda,t) + \Delta G(\lambda,t) \tag{2}$$

where ref denotes a time period without perturbations of the optical properties in the medium, considered as a reference period. We consider that $\Delta L(\lambda,t) = L\left(\mu_a,\mu_s\right) - L\left(\mu_a^{ref},\mu_s^{ref}\right)$.

The validity of Eq. (2) depends on how much L varies with the absorption and scattering coefficients. Using the Beer law, changes in the absorption coefficient can be linked to the concentration changes ΔC (in mol.L^{-1}) of the chromophore within the medium and its molar extinction coefficient ϵ_n (in L.mol^{-1}.cm^{-1}). Using a spectral acquisition device having K spectral channels, Eq. (2) can be expressed by the following matrix system:[3]

$$\begin{bmatrix} \Delta A_1(t) \\ \vdots \\ \Delta A_K(t) \end{bmatrix} = \begin{bmatrix} E_{1,1} & \dots & E_{1,N} \\ \vdots & \ddots & \vdots \\ E_{K,1} & \dots & E_{K,N} \end{bmatrix} \begin{bmatrix} \Delta C_1(t) \\ \vdots \\ \Delta C_N(t) \end{bmatrix} + \begin{bmatrix} B_1(t) \\ \vdots \\ B_K(t) \end{bmatrix} + \begin{bmatrix} \Delta G_1(t) \\ \vdots \\ \Delta G_K(t) \end{bmatrix} \tag{3}$$

The elements of the matrices E, B and G have been calculated as follows:

$$\begin{aligned} E_{i,n} &= \int_{\lambda_1}^{\lambda_2} \epsilon_n(\lambda).D_i(\lambda).S(\lambda).L\left(\mu_a^{ref},\mu_s^{ref}\right).d\lambda \\ B_i(t) &= \int_{\lambda_1}^{\lambda_2} D_i(\lambda).S(\lambda).\mu_a(\lambda,t).\Delta L(\lambda,t).d\lambda \\ \Delta G_i(t) &= \int_{\lambda_1}^{\lambda_2} D_i(\lambda).S(\lambda).\Delta G(\lambda,t).d\lambda \end{aligned} \tag{4}$$

The spectral sensitivity of the detector i of the spectral acquisition device is represented by $D_i(\lambda)$ and $S(\lambda)$ is the normalized intensity spectrum of the light source. $[\lambda_1; \lambda_2]$ is the wavelength range collected by the spectral acquisition device.

In functional brain mapping studies, it is commonly assumed that L does not vary with $\Delta\mu_a$ and $\Delta\mu_s$ (which is not strictly correct[7]). This implies that matrix B in Eq. (3) is null. In order to estimate ΔC values, elements of matrix E and ΔG need to be calculated. As cerebral activity is related to slow events,[8] it is common to apply a low-pass filtering on measured ΔA to remove the physiological noises such as heart rate and breathing rate. Elements of matrix ΔG depend on the geometry. These factors are mainly due to tissue dessication[9] that are estimated with linear regression. Once matrix ΔG has been removed from Eq. (3), we obtain Eq. (5):

$$[\Delta A(t)] = [E]\,[\Delta C(t)] \tag{5}$$

For the calculation of matrix E, we consider that the variations of absorbance due to the changes of absorption are mainly due to two chromophores ($N = 2$ in Eq. (3)), whereas these variations are in reality due to a very great unknown number of chromophores. In functional brain mapping studies based on RGB imaging, these chromophores are the oxy- and deoxygenated hemoglobin (HbO_2 and Hb). For calculating E, the optical path length $L\left(\mu_a^{ref}, \mu_s^{ref}\right)$ can be for example modeled using Monte Carlo simulations[3] or the diffusion approximation.[10] Finally, chromophores' concentration changes are obtained by matrix inversion.

2.3 Auto-calibration method

As we saw in section 2.2, the modified Beer-Lambert law is an approximate method based on assumptions that are not strictly correct. The principle of the auto-calibration method is to find a model that simplifies the expression of ΔA into two components (or sources) in a space where Eq. (5) is valid. This space can be found by studying the frequencies of ΔA, for which $\Delta\mu_a$ is very low and $\Delta\mu_s = 0cm^{-1}$. These conditions can be met if the concentration of chromophores varies weakly and periodically over time. Under these conditions, matrix B in Eq. (3) is null because $\Delta L(\lambda, t) \approx 0cm$. The factor ΔG in Eq. (3) may be approximated to 0 if it is evaluated in a narrow and high frequency bandwidth and not linked to periodic event, which leads us to Eq. (5).

For functional brain mapping, this space could be related to heart rate (narrow frequency band around 1 Hz) or breathing rate (narrow frequency band around 0.2 Hz). It is interesting to note that ΔA related to these physiological events are usually considered as noise and are discarded with a low-pass filtering. As absorption changes induced by respiratory or heartbeat are only due to Hb and HbO_2, it is valid to assume that only two chromophores vary ($N = 2$ in Eq. (3)). It is acceptable to assume that there is no change in μ_s since only concentrations of Hb and HbO_2 are periodically varying. These changes in μ_s are possibly induced by cell conformational changes and swelling[11] and are no larger than 0.4%[12] in the low frequency range. Although changes in μ_a are not null, but lower than those related to brain activity. μ_a varies much less than in the case of cerebral activity because the changes in blood flow and volume on the scale of the heartbeat or respiration are only effects of elasticity of the vessels which are thus weakly significant compared to the effects of the physiological cascade related to the cerebral hemodynamic response. Thus, in this space it is valid to assume that the matrix B in Eq. (3) is null. Finally, since we focus on frequencies higher than those related to the matrix ΔG, we can consider their elements as zero.

The auto-calibration method consisted of several steps. First, spectra of ΔA spectrum was computed for each spectral channel (magnitude of the Fourier transform). Negative frequencies of the centered FFT were discarded and only frequencies centered on the periodic event were kept. In this study, we only kept a narrow frequency band around the respiratory frequency ($0.23Hz \leq F \leq 0.34Hz$). These signals form the input of a rank-2 separable NMF which admits a computationally efficient closed-form solution[4] to estimate the matrix E in Eq. (5). ΔA signals were low-pass filtered (ΔA_{LP} of frequencies $F < 0.05Hz$[3]) to isolate the low-frequency modulation of light absorption linked to cortical activity.[8] Finally, we used the ΔA_{LP} signal to compute slow fluctuations of sources ΔC_{NMF} (in arbitrary units):

$$\Delta C_{NMF} = \Delta A_{LP} W^{+}, \tag{6}$$

with W the sources retrieved by the NMF algorithm and $^{+}$ the right pseudo-inverse operator.

2.4 Functional brain mapping

Pre-processing steps described in[3] are executed to compute absorbance changes time courses. The modified Beer-Lambert law (see section 2.2) and the auto-calibration method (see section 2.3) were used to measure hemodynamic responses in patient's brain during cerebral activity.

We applied the SPM technique combined with the random field theory[5,6] for the identification functional brain areas. Functional brain images computed with the modified Beer-Lambert law were compared to those computed with the auto-calibration method using the DICE coefficient.

3. RESULTS

Functional brain maps obtained with the modified Beer-Lambert law and the auto-calibration method are represented in Fig. 1. Statistical z-maps are plotted in false colors, green contours represent the extent of the functional brain area identified with SPM and the random field theory. The white point and the letter M indicate the motor area identified by electrical brain stimulation.

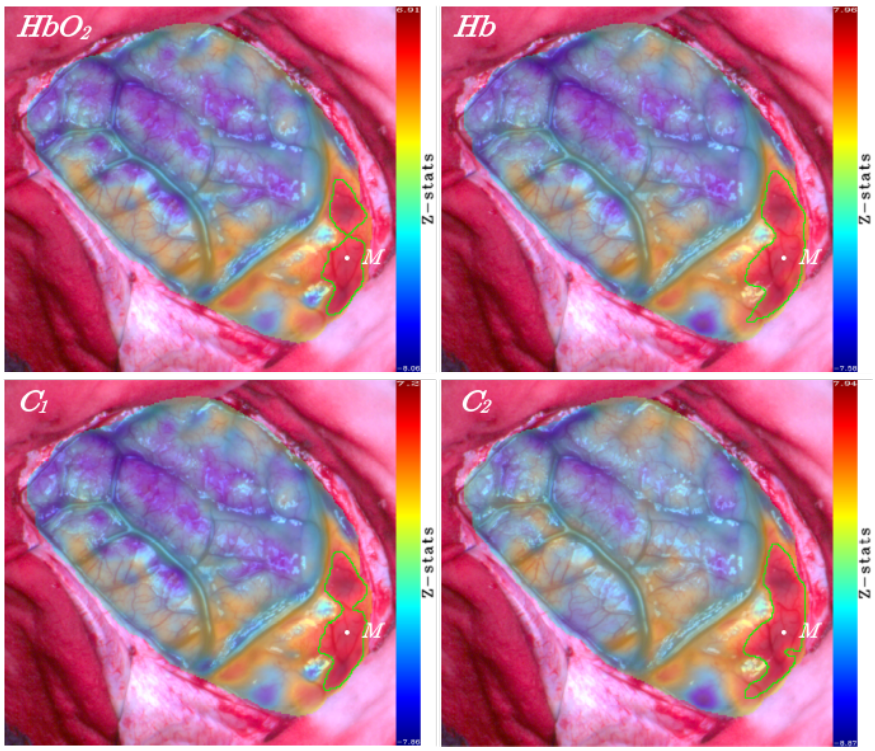

Figure 1. Functional brain maps obtained with the modified Beer-Lambert law and the auto-calibration method. Green contours represent the extent of the functional brain area identified with SPM and the random field theory. The white point and the letter M indicate the motor area identified by electrical brain stimulation.

When comparing SPM masks (binary masks delimited by the green contours), functional brain areas identified with auto-calibration method have a strong similarity with those identified with the modified Beer-Lambert law ($DICE(HbO_2, C_1) = 0.92$ and $DICE(Hb, C_2) = 0.93$). Functional brain maps identified with RGB imaging are consistent with identifications provided by electrical brain stimulation.

4. CONCLUSION

In conclusion, we proposed a new separable linear spectral unmixing method for the estimation of constituent spectra (endmembers) as well as their proportion in measurements: the auto-calibration method. This approach takes advantage of the periodic temporal variations of the sources in order to estimate them automatically

and without device calibration. We evaluated the performances of the method thought a clinical application: functional brain mapping. In the future, we plan to validate the method using a liquid and numerical phantom. This work can allow a widespread use of spectral imaging in the industrial or medical field. With for example, the use of camera smartphones in telemedicine applications in order to help patients and doctors to carry out a continuous functional monitoring of tissue oxygenation.

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